5 Interesting Reference Designs For USB-C POWER AND CHARGING

Explore USB-C reference designs for power delivery, battery charging, protection, and connections in electronic products.

NIDHI AGARWAL

If you are planning to add USB Type-C power or charging to your next design, you may already know that it involves more than just adding a USB-C connector. You also need to consider power negotiation, charging, protection, isolation, and thermal management. Here are some reference designs that can help you get started. You can use these reference designs when developing USB-C power inputs, USB power delivery chargers, multi-port charging systems, or isolated USB interfaces for consumer, industrial, automotive, and other applications.

USB Type-C power input

This reference design is intended for design engineers developing products that need



to use USB Type-C instead of a traditional DC barrel connector for power. You can use it as a starting point when designing USB Type-C power inputs for embedded systems, industrial equipment, consumer electronics, and accessories. The design uses the USB PD controller to manage power negotiation with a connected charger without needing a separate microcontroller. You can also change the connector placement, components, power path, and PCB size based on your requirements. The design includes protection against short-to-VBUS conditions and a status LED, allowing engineers to evaluate the power input and protection functions during development.

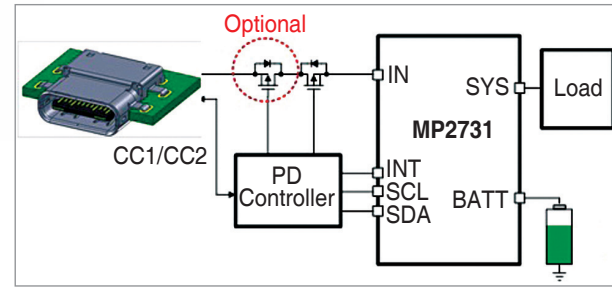
Applications. It makes it easier to switch to USB Type-C and can provide up to 100 watts of power.

Link. *https://efy.co.in/8xue3h*

OEM Brand. STMicroelectronics

USB Power Delivery battery charger

Monolithic Power Systems (MPS) has developed a USB Power Delivery reference design for charging single-cell lithium-ion batteries in smartphones, tablets, and other portable devices. The design follows USB PD 3.0 specifications and combines a high-current switching battery charger with a USB PD controller. It supports 5V to 16V input, up to 4.5A charging current, 18W USB PD fast charging, and more than 90 per cent efficiency. Its bidirectional USB Type-C port can charge the battery from a power source or provide 5V USB on-the-go power to another device. Protection features include battery-



temperature monitoring, thermal regulation, thermal shutdown, and input and system over-voltage protection.